

Reviewer Pack — Minimal Geometric Entanglement and Communication Complexity ...

Entry 46cbc77b2568 · INCONCLUSIVE

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Minimal Geometric Entanglement and Communication Complexity Rank Correlation

Entry ID: 46cbc77b2568 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-04 09:30:25 UTC

1. Conjecture

Field A (mathematical branch): Quantum Information Theory (Geometric Entanglement) **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For every n -variable Boolean function f , the minimal geometric entanglement ($\text{mge}(f)$) of its associated quantum circuit is linearly correlated with its communication complexity rank ($\text{ccr}(f)$), such that $\text{mge}(f) = \Omega(\text{ccr}(f))$.

Rationale (proposer's reasoning):

The field of quantum information theory has recently explored geometric entanglement as a measure of the difficulty in classifying quantum states. Correlating this with communication complexity rank could reveal new insights into the intrinsic complexity of Boolean functions.

Taxonomy category: Quantum Information Theory (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 7fbd58773b3be438

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For each n -variable Boolean function f , if the ratio between its minimal geometric entanglement ($\text{mge}(f)$) and communication complexity rank ($\text{ccr}(f)$) is greater than or equal to a predefined threshold (e.g., 2), then the conjecture is supported.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "quantum information theory" AND "geometric entanglement" AND "communication complexity" AND "matrix rank" - "entanglement" AND "communication complexity rank" AND "Boolean function" - "quantum circuit" AND "communication complexity" AND "minimal geometric entanglement"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=124, elapsed=240.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity_rank(f):
        n = int(math.log2(len(f)))
        if 2**n != len(f):
            return float('inf')

        # Simple heuristic: count the number of unique rows
        rows = set()
        for i in range(n + 1):
            row = [f[j] for j in range(2**i, min(2**(i+1), len(f)))]
            rows.add(tuple(row))

        return len(rows)

    def minimal_geometric_entanglement(f):
        n = int(math.log2(len(f)))
        if 2**n != len(f):
            return float('inf')

        # Simple heuristic: count the number of unique pairs
        pairs = set()
        for i in range(n + 1):
            for j in range(2**i, min(2**(i+1), len(f))):
                for k in range(j + 1, min(2**(i+1), len(f))):
```

```

        pair = (f[j], f[k])
        pairs.add(tuple(pair))

    return len(pairs)

n_values = [5, 10, 15, 20, 30, 40]
total_metric_value = 0
instances_tested = 0
n_max = 0

for n in n_values:
    f = generate_random_boolean_function(n)
    ccr_f = communication_complexity_rank(f)
    mge_f = minimal_geometric_entanglement(f)

    if ccr_f == float('inf') or mge_f == float('inf'):
        continue

    instances_tested += 1
    n_max = max(n_max, n)
    total_metric_value += mge_f / ccr_f

mean_metric_value = total_metric_value / instances_tested if instances_tested > 0 else 0
conjecture_holds = mean_metric_value >= 2
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": "mge_over_ccr",
    "metric_value": mean_metric_value,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}}, **result}}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results if r["instances_tested"] > 0) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot verify the conjecture's support or falsification. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14823
2	propose	ollama_remote	glm4:latest	0	0	15720
3	preregistration	ollama_remote	glm4:latest	0	0	9206
4	novelty	ollama_remote	glm4:latest	0	0	10706
5	novelty	ollama_remote	glm4:latest	0	0	11383
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17626
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7661
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14801
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10976
10	judge	ollama_remote	glm4:latest	0	0	26303

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 139206 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/46cbc77b2568.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/46cbc77b2568.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/46cbc77b2568.tar.gz` (if generated)